

QBIT Hub

Enterprise SaaS Portal

Executive Summary

The Enterprise Analytics module — Version 2 — has been successfully implemented, verified, and committed to GitHub. This report documents the architecture, feature inventory, verification results, and operational readiness.

The module delivers a business intelligence platform that transforms operational data into actionable insights. It aggregates real data from work orders, diagnostic sessions, device passports, downloads, warranty records, and knowledge base articles — across 10 analytics tabs (executive, device, engineer, service, customer, warranty, Dr. QBIT, download, branch, insights). The module NEVER fabricates predictions and only generates insights supported by collected data.

GitHub: [qbithubsoftware-sketch/qbithub](#) · Commit: 713d717 · Status: READY

Analytics

Tabs (10 sections)

Tab	Metrics Computed
Executive Dashboard	Total customers, installed devices, active devices, installations, relocations, service visits, open/completed work orders
Device Analytics	Most installed products/models/categories, most active/serviced devices, driver/firmware update stats
Engineer Performance	Jobs assigned/completed/pending, average completion hours, ratings, photo compliance, report submission rate
Service Analytics	Most common service requests, installation problems, driver/firmware issues, most viewed KB articles
Customer Analytics	Customer-level aggregation (from existing customer + work order data)
Warranty Analytics	Active, expired, expiring in 30/60/90 days
Dr. QBIT Analytics	Total/successful/failed scans, unknown devices, driver/firmware mismatches, common findings
Download Analytics	Most downloaded drivers/manuals/firmware, most opened KB articles
Branch Analytics	Branch comparison: devices, completed jobs, open jobs by branch/city/state
Insights	Evidence-based insights: highest service frequency, fastest engineer, warranty expiring, driver updates needed

Document

Enterprise Analytics Readiness Report

Module Version: 2.0 | Date: 14 July 2026

Audience: Administrators

3. Architecture

Model	Purpose
AnalyticsSnapshot	Periodic snapshot of key metrics (JSON blob per type). For trend analysis.
BusinessMetric	Individual KPI with trend tracking (current vs previous value).
EngineerMetric	Per-engineer performance: jobs, ratings, completion time, compliance.
DeviceMetric	Per-device health/activity: scans, tests, diagnostics, failures.

3.1 API Endpoints (11 new)

Endpoint	Purpose
GET /api/analytics/executive	8 executive KPIs (customers, devices, work orders)
GET /api/analytics/device	Device analytics (8 metrics + top lists)
GET /api/analytics/engineer	Per-engineer performance metrics
GET /api/analytics/service	Common issues by category + KB views
GET /api/analytics/warranty	Warranty distribution (active/expired/expiring)
GET /api/analytics/dr-qbit	Scan stats + diagnostic findings
GET /api/analytics/download	Most downloaded/viewed
GET /api/analytics/branch	Branch comparison table
GET /api/analytics/insights	Evidence-based insights
POST/GET /api/analytics/reports	Generate + list analytics reports

4. Verification Results

Check	Result	Notes
TypeScript (tsc --noEmit)	PASS	Zero errors across all analytics files
Next.js Production Build	PASS	10 new analytics routes registered
ESLint	PASS	No new lint errors
Git Push	DONE	Commit 713d717 on main

4.1 Analytics Accuracy Verification

- All metrics computed from real Prisma queries on existing tables ✓
- No fabricated predictions or placeholder statistics ✓
- Executive KPIs: direct count() queries on WorkOrder, FSMCustomer, DevicePassport ✓
- Device analytics: aggregation from DevicePassport + DriverInfo + FirmwareInfo ✓
- Engineer performance: from WorkOrder + InstallationRating + completedAt timestamps ✓
- Warranty: from DeviceWarranty with date range queries ✓
- Dr. QBIT: from ScanSession + DiagnosticFinding counts ✓
- Download: from Download.downloadCount + KnowledgeArticle.viewCount ✓

4.2 RBAC Verification

- Analytics screen is admin-only (RBAC: administrator only) ✓
- All API endpoints use requireAuth() ✓
- Customers view only their own info (via public tracking page) ✓
- Engineers view only assigned info (via FSM RBAC) ✓

4.3 Insights Verification (Evidence-Based)

- Insight 1: Product with highest service frequency — from WorkOrder groupBy assetId ✓
- Insight 2: Engineer with fastest completion — from completed WO durations ✓
- Insight 3: Warranty expiring within 90 days — from DeviceWarranty date queries ✓
- Insight 4: Driver updates needed — from DriverInformation count ✓
- Every insight includes evidence string (query description) ✓

4.4 Responsive Layout Verification

- Tab navigation: horizontal scroll on mobile ✓
- KPI cards: 2 cols mobile, 4 cols desktop ✓
- Top lists: 1 col mobile, 2 cols desktop ✓
- Engineer table: horizontal scroll on mobile ✓

4.5 TypeScript Verification

- tsc --noEmit passes with 0 errors ✓
- All types strictly defined: AnalyticsTab, ExecutiveDashboardDTO, DeviceAnalyticsDTO, EngineerAnalyticsDTO, etc. ✓

5. Reused Existing Modules

- WorkOrder + FSMCustomer + FSMCustomerAsset — service/customer data
- DevicePassport + QbitProduct — device data
- DriverInformation + FirmwareInformation + DeviceWarranty — health data
- DiagnosticSession + DiagnosticFinding — diagnostics data
- ScanSession + DetectedDevice + UnknownDevice — detection data
- Download + DownloadHistory + KnowledgeArticle — download/view data
- Branch — branch locations (from Fleet module)
- InstallationRating — engineer ratings
- All primitives — Icon, SurfaceCard, QbitButton, StatusBadge, TagBadge

6. Security

Analytics visibility respects RBAC. Customers view only their own information. Engineers view only assigned information. Administrators have complete analytics. The analytics screen is admin-only in the RBAC matrix. All API endpoints use `requireAuth()` which returns 401 for unauthenticated requests.

7. Conclusion

The Enterprise Analytics module — Version 2 — is fully implemented, verified, and committed to GitHub. The module provides 10 analytics tabs with real data aggregated from all existing modules — no fabricated predictions. Evidence-based insights are generated only from collected data, with transparent evidence strings showing exactly which queries produced each insight.

Status: READY FOR PRODUCTION · Verified: 14 July 2026 · Owner: QBIT Hub Engineering Team

Next module awaiting approval: Final Production Readiness Audit — as specified in the brief, implementation will begin only after explicit approval.